

Newman-Penrose Formalism Calculations

Stephani et al., Equation (31.34)

Metric

$$ds^2 = \left(- \left(\sqrt{2} \frac{\partial W_x}{\partial x} - \sqrt{2} \frac{\partial W_y}{\partial y} \right) v - 2 H_0 \right) du \otimes du - du \otimes dv - \sqrt{2} W_x du \otimes dx + \sqrt{2} W_y du \otimes dy - dv \otimes du - \sqrt{2} W_x dx \otimes du + dx \otimes dx + \sqrt{2} W_y dy \otimes du + dy \otimes dy$$

Coordinates

$$(x^\mu) = (u, v, x, y)$$

Metric Functions

$$W_x(u, x, y) = W_x$$

$$H(u, v, x, y) = H$$

$$W_y(u, x, y) = W_y$$

Null Tetrad

Covariant components

$$l = (1) du$$

$$n = \left(\frac{1}{2} \sqrt{2} v \frac{\partial}{\partial x} W_x - \frac{1}{2} \sqrt{2} v \frac{\partial}{\partial y} W_y + H_0 \right) du + (1) dv + \left(\sqrt{2} W_x \right) dx + \left(-\sqrt{2} W_y \right) dy$$

$$m = \left(\frac{1}{2} \sqrt{2} \right) dx + \left(\frac{1}{2} i \sqrt{2} \right) dy$$

$$\overline{m} = \left(\frac{1}{2} \sqrt{2} \right) dx + \left(-\frac{1}{2} i \sqrt{2} \right) dy$$

Contravariant components

$$l = (-1) dv$$

$$n = (-1) du + \left(\frac{1}{2} \sqrt{2} v \frac{\partial}{\partial x} W_x - \frac{1}{2} \sqrt{2} v \frac{\partial}{\partial y} W_y + H_0 \right) dv$$

$$m = (-W_x + i W_y) dv + \left(\frac{1}{2} \sqrt{2} \right) dx + \left(\frac{1}{2} i \sqrt{2} \right) dy$$

$$\overline{m} = (-W_x - i W_y) dv + \left(\frac{1}{2} \sqrt{2} \right) dx + \left(-\frac{1}{2} i \sqrt{2} \right) dy$$

Directional Derivatives

$$D = -\partial_v$$

$$\Delta = -\partial_u + \frac{1}{2} \sqrt{2} v \frac{\partial}{\partial x} W_x - \frac{1}{2} \sqrt{2} v \frac{\partial}{\partial y} W_y + H_0 \partial_v$$

$$\delta = -W_x + i W_y \partial_v + \frac{1}{2} \sqrt{2} \partial_x + \frac{1}{2} i \sqrt{2} \partial_y$$

$$\overline{\delta} = -W_x - i W_y \partial_v + \frac{1}{2} \sqrt{2} \partial_x - \frac{1}{2} i \sqrt{2} \partial_y$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{1}{2}i\sqrt{2}\frac{\partial}{\partial y}W_x + \frac{1}{2}i\sqrt{2}\frac{\partial}{\partial x}W_y$$

$$\lambda = 0$$

$$\begin{aligned} \nu = & -\frac{1}{2}\sqrt{2}W_x\frac{\partial}{\partial x}W_x - \frac{1}{2}i\sqrt{2}W_y\frac{\partial}{\partial x}W_x + \frac{1}{2}\sqrt{2}W_x\frac{\partial}{\partial y}W_y + \frac{1}{2}i\sqrt{2}W_y\frac{\partial}{\partial y}W_y + \frac{1}{2}v\frac{\partial^2}{(\partial x)^2}W_x - \frac{1}{2}i v\frac{\partial^2}{\partial x\partial y}W_x \\ & - \frac{1}{2}v\frac{\partial^2}{\partial x\partial y}W_y + \frac{1}{2}i v\frac{\partial^2}{(\partial y)^2}W_y + \frac{1}{2}\sqrt{2}\frac{\partial}{\partial x}H_0 - \frac{1}{2}i\sqrt{2}\frac{\partial}{\partial y}H_0 - \frac{\partial}{\partial u}W_x - i\frac{\partial}{\partial u}W_y \end{aligned}$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = -\frac{1}{4}\sqrt{2}\frac{\partial}{\partial x}W_x + \frac{1}{4}i\sqrt{2}\frac{\partial}{\partial y}W_x + \frac{1}{4}i\sqrt{2}\frac{\partial}{\partial x}W_y + \frac{1}{4}\sqrt{2}\frac{\partial}{\partial y}W_y$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = 0$$

$$\Phi_{12} = -\frac{1}{4} \frac{\partial^2}{(\partial x)^2} W_x - \frac{1}{4} \frac{\partial^2}{(\partial y)^2} W_x + \frac{1}{4} i \frac{\partial^2}{(\partial x)^2} W_y + \frac{1}{4} i \frac{\partial^2}{(\partial y)^2} W_y$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = -\frac{1}{4} \frac{\partial^2}{(\partial x)^2} W_x - \frac{1}{4} \frac{\partial^2}{(\partial y)^2} W_x - \frac{1}{4} i \frac{\partial^2}{(\partial x)^2} W_y - \frac{1}{4} i \frac{\partial^2}{(\partial y)^2} W_y$$

$$\begin{aligned} \Phi_{22} = & \frac{1}{4} \sqrt{2} v \frac{\partial^3}{(\partial x)^3} W_x + \frac{1}{4} \sqrt{2} v \frac{\partial^3}{\partial x (\partial y)^2} W_x - \frac{1}{4} \sqrt{2} v \frac{\partial^3}{(\partial x)^2 \partial y} W_y - \frac{1}{4} \sqrt{2} v \frac{\partial^3}{(\partial y)^3} W_y - \frac{1}{2} \frac{\partial}{\partial x} W_x^2 - W_x \frac{\partial^2}{(\partial x)^2} W_x + W_y \frac{\partial^2}{\partial x \partial y} W_x + \frac{1}{2} \frac{\partial}{\partial y} W_x^2 + \frac{\partial}{\partial y} W_x \frac{\partial}{\partial x} W_y \\ & + \frac{1}{2} \frac{\partial}{\partial x} W_y^2 + W_x \frac{\partial^2}{\partial x \partial y} W_y + \frac{\partial}{\partial x} W_x \frac{\partial}{\partial y} W_y - \frac{1}{2} \frac{\partial}{\partial y} W_y^2 - W_y \frac{\partial^2}{(\partial y)^2} W_y - \frac{1}{2} \sqrt{2} \frac{\partial^2}{\partial u \partial x} W_x + \frac{1}{2} \sqrt{2} \frac{\partial^2}{\partial u \partial y} W_y + \frac{1}{2} \frac{\partial^2}{(\partial x)^2} H_0 + \frac{1}{2} \frac{\partial^2}{(\partial y)^2} H_0 \end{aligned}$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = 0$$

$$\Psi_3 = -\frac{1}{4} \frac{\partial^2}{(\partial x)^2} W_x + \frac{1}{2} i \frac{\partial^2}{\partial x \partial y} W_x + \frac{1}{4} \frac{\partial^2}{(\partial y)^2} W_x + \frac{1}{4} i \frac{\partial^2}{(\partial x)^2} W_y + \frac{1}{2} \frac{\partial^2}{\partial x \partial y} W_y - \frac{1}{4} i \frac{\partial^2}{(\partial y)^2} W_y$$

$$\begin{aligned} \Psi_4 = & \frac{1}{4} \sqrt{2} v \frac{\partial^3}{(\partial x)^3} W_x - \frac{1}{2} i \sqrt{2} v \frac{\partial^3}{(\partial x)^2 \partial y} W_x - \frac{1}{4} \sqrt{2} v \frac{\partial^3}{\partial x (\partial y)^2} W_x - \frac{1}{4} \sqrt{2} v \frac{\partial^3}{(\partial x)^2 \partial y} W_y + \frac{1}{2} i \sqrt{2} v \frac{\partial^3}{\partial x (\partial y)^2} W_y + \frac{1}{4} \sqrt{2} v \frac{\partial^3}{(\partial y)^3} W_y - \frac{1}{2} \frac{\partial}{\partial x} W_x^2 - W_x \frac{\partial^2}{(\partial x)^2} W_x \\ & - i W_y \frac{\partial^2}{(\partial x)^2} W_x + i W_x \frac{\partial^2}{\partial x \partial y} W_x - W_y \frac{\partial^2}{\partial x \partial y} W_x + \frac{1}{2} i \frac{\partial}{\partial x} W_x \frac{\partial}{\partial y} W_x - \frac{1}{2} i \frac{\partial}{\partial x} W_x \frac{\partial}{\partial x} W_y + W_x \frac{\partial^2}{\partial x \partial y} W_y + i W_y \frac{\partial^2}{\partial x \partial y} W_y - \frac{1}{2} i \frac{\partial}{\partial y} W_x \frac{\partial}{\partial y} W_y + \frac{1}{2} i \frac{\partial}{\partial x} W_y \frac{\partial}{\partial y} W_y \\ & + \frac{1}{2} \frac{\partial}{\partial y} W_y^2 - i W_x \frac{\partial^2}{(\partial y)^2} W_y + W_y \frac{\partial^2}{(\partial y)^2} W_y - \frac{1}{2} \sqrt{2} \frac{\partial^2}{\partial u \partial x} W_x + \frac{1}{2} i \sqrt{2} \frac{\partial^2}{\partial u \partial y} W_x - \frac{1}{2} i \sqrt{2} \frac{\partial^2}{\partial u \partial x} W_y - \frac{1}{2} \sqrt{2} \frac{\partial^2}{\partial u \partial y} W_y + \frac{1}{2} \frac{\partial^2}{(\partial x)^2} H_0 - i \frac{\partial^2}{\partial x \partial y} H_0 - \frac{1}{2} \frac{\partial^2}{(\partial y)^2} H_0 \end{aligned}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "III"